

AYUSH KUMAR JHA

B.E. Electronics & Communication Engineering (VTU) • Cybersecurity Minor (IIT Guwahati) • Embedded Systems & Applied AI

Bengaluru, India • +91 75015 63105 • ayushjha.kviitkgp@gmail.com
linkedin.com/in/ayush-kumarjha • github.com/AyushKJha



Profile

Second-year Electronics & Communication Engineering student with interests in cybersecurity, embedded systems, computer vision, and full-stack development. Seeking software engineering or research internships to build secure, intelligent, and scalable systems.

Experience

Social Media Manager | Technical Events Coordinator— TechHub-Official Technical Community, SMVIT
Bengaluru, Karnataka, India Sep 2025 – Present

- Spearheaded the digital outreach and social media strategy for a college developer community, creating technical content on cybersecurity, AI, embedded systems, and software development.
- Increased the community's Instagram following by 1000+ followers within four hours through a high-impact promotional campaign, significantly improving online reach and engagement.
- Served as the Cybersecurity Lead and Mentor for TechVistara 5.0, mentoring first-year students and leading cybersecurity-focused learning sessions and activities.
- Managed digital promotions, sponsor coordination, and technical operations for HackNocturne 2.0, collaborating with organising teams to ensure smooth execution of the hackathon.
- Co-organised Code Café, a campus coding hackathon, along with multiple technical seminars and webinars, handling event promotions, participant engagement, and cross-functional coordination.
- Collaborated with design, technical, and organising teams to deliver consistent branding, promotional content, and successful execution of community events.

Projects

Daily Routine Tracker | Live Demo 2025
Python • SQLite • JWT • Claude AI API

- Multi-user habit tracker: 15+ REST endpoints, PBKDF2-SHA256 password hashing, JWT sessions (7-day expiry), and server-side rate limiting.
- Integrated the Claude AI API for personalised coaching; 3-table SQLite schema; deployed via Nginx on a Linux VPS, Render, Railway, and Fly.io, with 30+ analytics visualisations.

Education

Visvesvaraya Technological University
B.E. — Electronics & Communication Engineering
Sep 2024 – Jul 2028 (Expected)
CGPA: 8.8/10 (4th Semester)

IIT Guwahati
Minor Program — Cybersecurity
2024 – 2025 • Score: 85%

Kendriya Vidyalaya, IIT Kharagpur
Class XII (CBSE, Science with CS): 87% (2023)
Class X (CBSE): 96% (2021)

Technical Skills

Languages

Python C++
JavaScript HTML/CSS

Cybersecurity

Nmap Wireshark
Burp Suite Kali Linux

OWASP Top 10

JWT PBKDF2

Embedded & IoT

ESP32 Arduino
Bluetooth PWM
Sensors

Full-Stack & Cloud

Node.js SQLite
REST APIs Nginx
Git Linux

Computer Vision & DSA

OpenCV MediaPipe
Data Structures
Algorithms

Cybersecurity Threat Lab

2025

Nmap • Wireshark • Kali Linux • NIST/ISO 27001

- Network reconnaissance and OWASP Top 10 vulnerability assessments using Nmap and Wireshark; simulated SOC threat-detection workflows on Kali Linux.
- Authored a four-volume compliance audit mapping organisational controls to the NIST CSF and ISO/IEC 27001 frameworks.

Android-Controlled ESP32 Smart Car | Demo Video

2025

C++ • ESP32 • Bluetooth • Arduino IDE

- Bluetooth-controlled robotic car: Android app control, ultrasonic obstacle avoidance, 360-degree range detection, and PWM motor control via an L298N driver across six commits.

Real-Time Hand Gesture Recognition System | Demo Video

2025

Python • OpenCV • MediaPipe

- Real-time hand-gesture recognition using MediaPipe’s 21-point landmark model with custom finger-state classification, processing 25+ FPS on standard hardware.

Research Interests

- Cybersecurity
- Full-Stack Dev
- Embedded/IoT
- Computer Vision
- DSA
- Cloud/DevSecOps

Achievements and Certifications

- **IIT Guwahati Cybersecurity Minor:** Cryptography, network defence, threat analysis, compliance frameworks. Score: **85%**.
- **IIT Khragpur Cloud Computing:** Cloud architecture and deployment models. Score: **77%**.
- **Four-Volume Cybersecurity Research & Lab Series:** Self-published security assessments (Nmap/Wireshark, OWASP, NIST/ISO 27001).
- **GitHub Pro:** Active contributor with four public projects across full-stack, IoT, computer vision, and cybersecurity.